

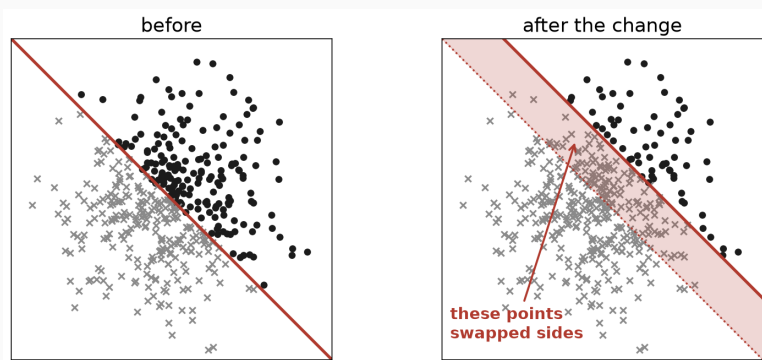
The Blind Spot Paradox

When a self-repairing model destroys the evidence its own monitor needs

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Concept drift: the rule changes, the data does not



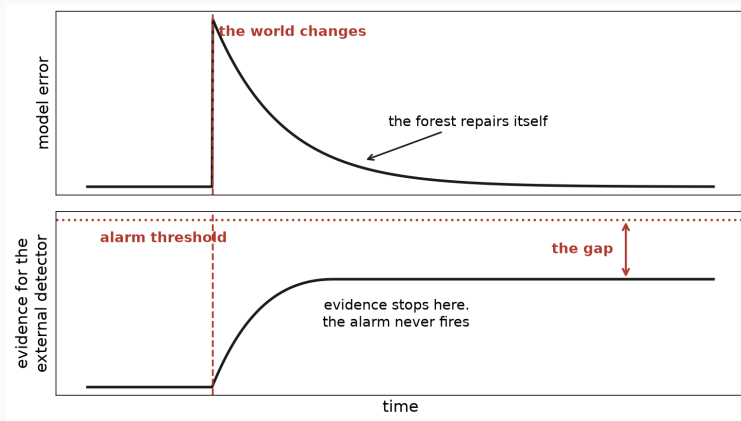
A model trained on the left keeps predicting, confidently, and is now **wrong on the shaded band**.

Two defences, built by two separate communities

- **A model that repairs itself.** A forest of ten small decision trees. When a tree starts failing, it is thrown away and a fresh one takes its place.
- **A watchdog that raises the alarm.** It reads the model's mistakes, adds up the excess, and calls for a human once the pile crosses a threshold.

The alarm is the only thing a human ever sees.

Put them together, and they cancel out



schematic

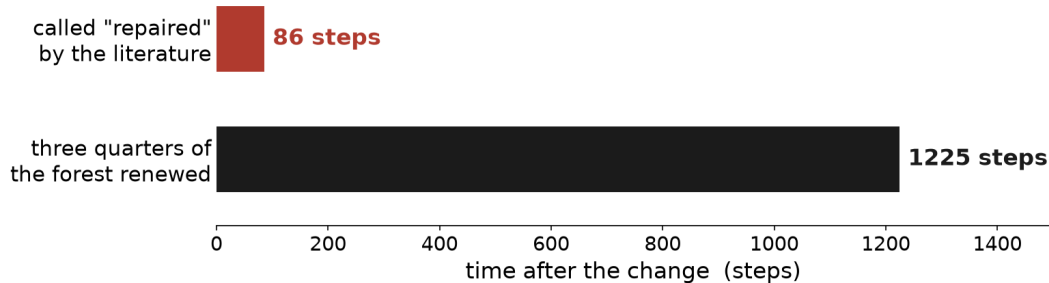
How often? We ran the experiment 2 000 times

we move the boundary at an instant we choose, then watch whether the alarm ever fires

it fired 3 times
out of 2 000 runs

- One community measures how fast a model recovers. The other measures how fast an alarm fires.
- Neither had measured them against each other.

Our contribution: the accepted clock is wrong



median over 100 runs at $\Delta e = 0.24$, $M = 10$

It marks the first tree out of ten, not a repaired forest.

Where we stand

- The paradox is **real and reproducible**, and we can say when the evidence falls short of what the alarm needs.
- The clock the field relies on is **biased, always the same way**. Every conclusion drawn from it inherits that bias.
- **Next:** watch the model and the watchdog on one common clock, and turn this into advice a practitioner can use.

Thank you.